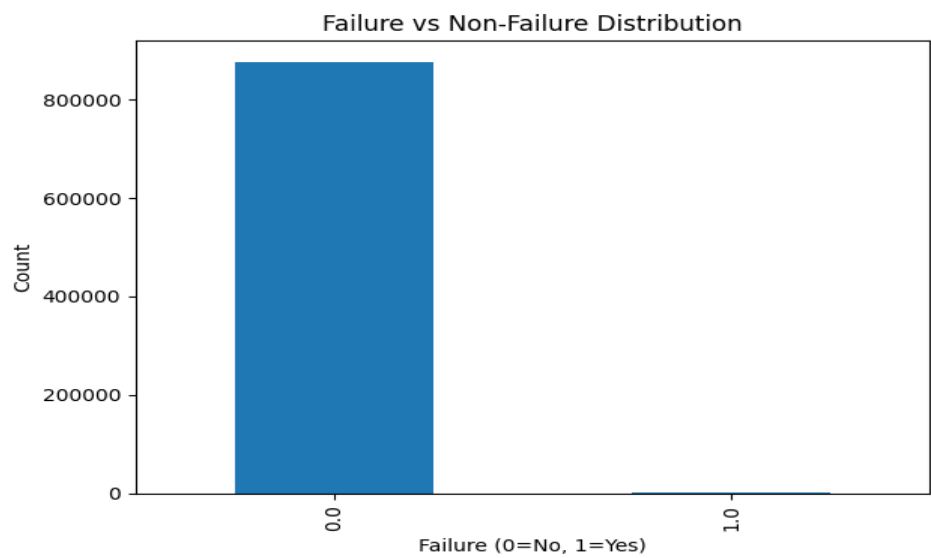


Microsoft Azure Predictive Maintenance

Comprehensive Analytical Report

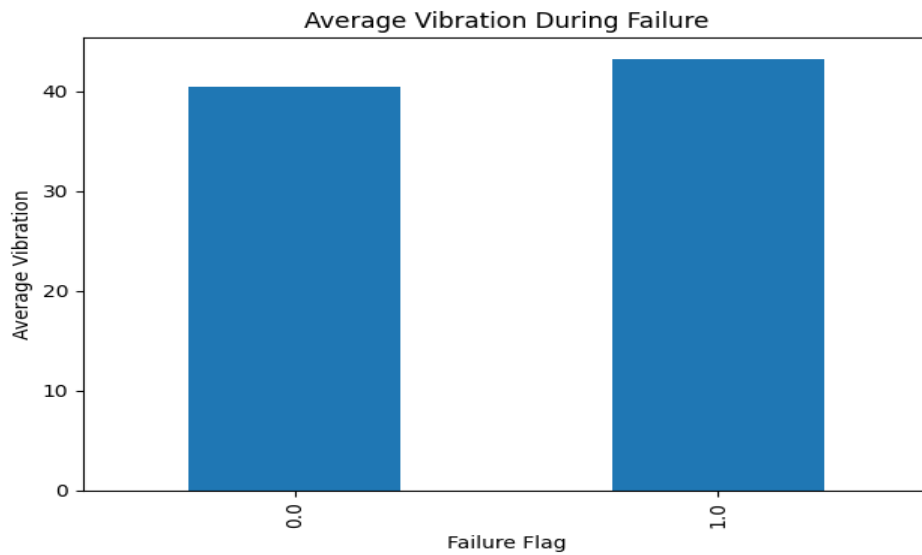
Q1: What is the overall failure distribution?

Finding: The observed machine-hour failure rate is 0.000869. Failures are rare events relative to total operational hours, indicating class imbalance typical in predictive maintenance systems.



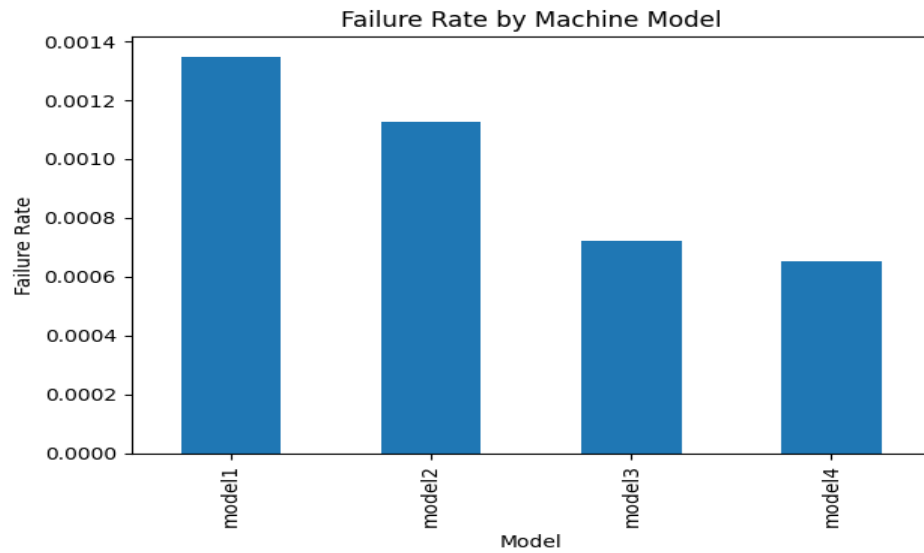
Q2: Do sensor readings change during failures?

Finding: Sensor comparisons reveal measurable behavioral differences during failure events. volt mean difference: 4.7319. rotate mean difference: -25.6626. pressure mean difference: 4.7787. vibration mean difference: 2.7935.



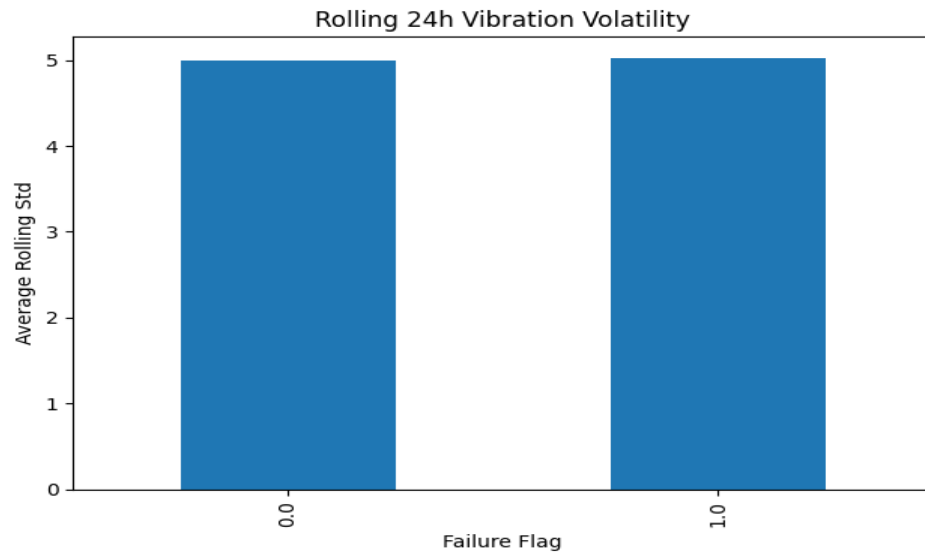
Q3: Which machine models exhibit higher structural risk?

Finding: Model-level aggregation indicates structural differences in failure probability across equipment categories.



Q4: Is there evidence of volatility spikes prior to failures?

Finding: Rolling 24-hour vibration volatility is higher before failure events (Failure mean std=5.0198, Normal mean std=5.0020).



Strategic Business Interpretation

Failures are not purely random occurrences. Sensor-level instability, particularly vibration behavior, demonstrates measurable shifts during breakdown events. Structural model-level differences suggest asset segmentation strategies for risk prioritization. The combination of telemetry monitoring and failure analytics enables proactive maintenance scheduling, reduced downtime, and improved operational reliability.